

HMIS DIMENSIONAL DATA MODEL

Data Model Usage Guide

How to read the model, choose the right table, and build reports that answer the real business question

Kimball Dimensional Model

Power BI

DAX

Hospital Analytics

Prepared by Rahuldev Veludandi

A companion manual to the HMIS Dimensional Data Modeling Project Report

PART 1

Part 1 — Read Before You Build

Focus *Before building anything, how should an analyst understand this model and decide where the answer should come from?*

Purpose and Intended Users

The HMIS dimensional model is designed as a hospital business intelligence model that can support analysis across multiple hospital functions rather than serving a single department. The model therefore acts as a common analytical layer for hospital business intelligence.

Stakeholder	Typical Use
Hospital Management	Overall hospital performance and cross-functional analysis
Operations	Admissions, departments, wards, beds and operational activity
Finance	Billing, payments and insurance-covered amounts
Clinical / Medical Teams	Diseases, diagnostics, doctors and clinical activity
Insurance Team	Insurance providers, policies and coverage
Pharmacy / Inventory	Drugs, stock levels and reorder conditions
HR / Staff Management	Employees, doctors, ward assignments and shifts
BI / Data Analysts	Dashboard development, reporting and deeper analysis

Read the Model Before Building Reports

KEY USAGE PRINCIPLE

The analyst must understand the data model before creating calculations, visuals or dashboards.

Before using the model, the analyst should study:

- The purpose of each table
- The grain of each table
- Fact and dimension classification
- Cardinality of relationships
- Filter direction
- Active and inactive relationships

- Which dimensions connect to which facts
- The purpose of dim_admission
- The role of snapshot_admission
- Available standardized DAX measures

This matters because the model contains several business processes with different grains. Billing is represented at bill grain, diagnostics at diagnostic-event grain, prescriptions at prescription grain, while snapshot_admission represents the admission journey at one row per admission. Selecting a table simply because it contains a related field or number can therefore produce an incorrect business result.

Business Question → Data Selection Framework

Before creating a visual or writing DAX, follow this eight-step sequence:

- 1 Identify the business question.**
Clearly define what needs to be answered. Example: "How much did the hospital bill in FY2025?"
- 2 Identify the business process.**
Determine which hospital process the question belongs to (e.g. Billing).
- 3 Identify the required grain.**
Determine what one analytical record represents (e.g. one bill).
- 4 Select the appropriate fact / dimension.**
e.g. fact_billing
- 5 Use the standard measure if available.**
e.g. Total Billing Amount
- 6 Identify the correct date.**
e.g. bill_date
- 7 Apply relevant dimensions / filters.**
e.g. Fiscal Year from dim_date
- 8 Validate the result.**
Confirm that the calculation represents the original business question.

The overall workflow:

Business Question → Process → Grain → Fact / Dimension → Standard Measure → Date / Date Role → Filters → Validate → Build Report

This approach helps prevent a common BI problem: a calculation can be technically correct but still answer the wrong business question.

KPI and Fact Selection Guide

The model contains separate facts for different hospital processes. The analyst should select the fact according to the business question.

Business Requirement	Primary Table	Typical Information
Billing amount	fact_billing	Total billing, insurance covered, patient payable
Bill count	fact_billing	Number of bills
Admissions	snapshot_admission	Admission-level journey
Admission attributes	dim_admission	Admission type, status and admission context
Diagnostic activity	fact_patient_diagnostics	Diagnostic events, results, doctors and tests
Prescription activity	fact_prescription	Drugs, dosage, frequency and duration
Current inventory	fact_drug_inventory	Current stock, reorder level and inventory status
Insurance policies	fact_patient_insurance	Policy and coverage information
Staff assignments	fact_staff_assignment	Employee-to-ward assignment and shift
Admission journey / LOS	snapshot_admission	Admission and discharge-related journey information

These are guidelines based on the intended business use of the model. The analyst should still examine the model and business question before selecting a table.

Admission-Specific Usage Rules

Admission analysis requires particular attention because both dim_admission and snapshot_admission are available for different analytical purposes.

Requirement	Recommended Source	Approach
Total number of admissions	snapshot_admission	Count admission records

Requirement	Recommended Source	Approach
Active patient / admission analysis	dim_admission	Use admission dimension
Admissions by admission type	dim_admission	Group / filter by admission type
Admissions by disease	dim_disease → dim_admission	Filter admission context through disease
Admissions by department	dim_beds → dim_admission	Filter admission context through bed/department
Admissions by year	dim_date → snapshot_admission	Use active admission-date relationship
Length of stay	snapshot_admission	Use admission journey information

"Admissions" is not a single universal calculation for every business question — the correct source depends on what the analyst is trying to measure.

For example, the model's total-admission KPI and an admission-dimension analysis may use different tables because they answer slightly different analytical questions.

Standardized KPI Measures

A set of standard DAX measures has been created and stored separately in an empty measure table. Examples include:

- Total Billing Amount
- Total Bills
- Total Admissions
- Diagnostic Events
- Insurance Covered
- Total Patients
- Total Prescriptions
- Other standard business measures

The purpose of these measures is to establish consistent KPI definitions across different users and departments. For example, if two departments require total billing, both should use the same standardized Total Billing Amount measure rather than independently creating their own versions.

USAGE PRINCIPLE

Use an existing standardized measure whenever the required KPI is already available. Create new DAX only when the business requirement genuinely requires a new calculation.

Part 1 is therefore about one central idea: before touching Power BI visuals or DAX, understand the model and determine exactly where the answer should come from.

PART 2

Part 2 — Use It the Right Way

Now that the analyst knows where the data comes from, how should relationships, dates, DAX and filters be used correctly?

Filter and Relationship Usage

The model should be used according to its defined relationship structure. Before creating a visual or calculation, the analyst should inspect the dimension-to-fact relationships, cardinality, filter direction, and active/inactive status.

Use the existing relationship path first. Use DAX only when the business requirement or relationship configuration requires it.

For example:

- dim_drugs → fact_prescription / fact_drug_inventory
- dim_doctors → fact_patient_diagnostics
- dim_diseases → dim_admission
- dim_beds → dim_admission
- dim_patients → admission-related analysis
- dim_insurance_provider → fact_patient_insurance

Not every dimension is intended to filter every fact. For example, filtering a drug should affect prescription or inventory analysis, but it should not be expected to change unrelated hospital admission or billing measures.

A KPI not changing under a particular slicer is not automatically a model error. The expected behaviour depends on the relationship path and the business question.

Date and DAX Usage

dim_date is the common date dimension for the processes where an appropriate source date exists. The current date relationships include:

Table	Date Column	Relationship
<i>fact_billing</i>	bill_date	Active
<i>fact_patient_diagnostics</i>	test_date	Active
<i>snapshot_admission</i>	admission_date	Active
<i>snapshot_admission</i>	discharge_date	Inactive
<i>fact_drug_inventory</i>	last_restock_date	Active

Calendar Year vs Fiscal Year

The model provides both a Calendar Year (January to December) and a Fiscal Year used for business/accounting analysis. The analyst should select the appropriate one based on the business requirement.

- "Billing by financial year" — Use Fiscal Year.
- "Admissions by calendar year" — Use Calendar Year with the appropriate admission date.

Inactive Date Relationship

discharge_date is an inactive relationship to dim_date. Therefore, a normal date slicer does not automatically use discharge date for analysis. If the business question is "How many admissions were discharged in each year?", the analyst must explicitly use the discharge-date relationship through appropriate DAX rather than assuming the active admission-date relationship answers the question.

Business Examples / Case Studies

CASE 1 · FINANCE — BILLING BY FISCAL YEAR

"What was the hospital's total billing amount by fiscal year?"

Fact: fact_billing

Measure: Total Billing Amount

Time/Date Context: dim_date

Relationship path: dim_date → fact_billing

Time attribute: Fiscal Year

The analyst should not calculate official billing totals from billing_detail.

CASE 2 · OPERATIONS — ADMISSIONS BY DEPARTMENT

"How many admissions were associated with each department?"

Admission context: dim_admission

Department: dim_beds

Relationship path: dim_beds → dim_admission

Admission count: appropriate admission calculation

The relationship path should be checked before building the visual.

CASE 3 · CLINICAL MANAGEMENT — DIAGNOSTIC ACTIVITY

“Which departments or doctors are handling more diagnostic activity?”

Fact: fact_patient_diagnostics

Measure: Diagnostic Events

Doctor: dim_doctors

Diagnostic test: dim_diagnostic_tests

Time/Date: dim_date

Relationship path: dim_doctors → fact_patient_diagnostics
 dim_diagnostic_tests → fact_patient_diagnostics
 dim_date → fact_patient_diagnostics

This type of analysis can also support resource-management decisions, such as identifying areas with higher diagnostic workload.

CASE 4 · PHARMACY — LOW INVENTORY

“Which drugs currently require attention?”

Fact: fact_drug_inventory

Dimension: dim_drugs

Attributes: current_stock, reorder_level, inventory_status

Filter: Low inventory

This represents the current inventory state. It should not be interpreted as historical stock movement because the source does not contain inventory movement history.

CASE 5 · MANAGEMENT — PATIENT LENGTH OF STAY

“How long are patients staying in the hospital?”

Table: snapshot_admission (one row per admission)

Measure / attribute: Length of Stay

Relevant dimensions: Patient, disease, bed/department, admission attributes

Date: Admission Date for admission-based analysis

If the question instead concerns discharge timing, the analyst must use the inactive discharge-date relationship appropriately.

Important Usage Cautions

- **Do not create fact-to-fact relationships** — The model was deliberately designed to avoid direct fact-to-fact relationships. Facts should be analysed through their appropriate dimensions and relationship paths.
- **Do not expect every slicer to affect every KPI** — A dimension should only affect a fact when an appropriate relationship path exists — for example, Drug → Prescriptions/Inventory does not mean Drug → Admissions/Billing. This is intentional model behaviour.
- **Use fact_billing for official billing measures** — The official financial measures should come from fact_billing. billing_detail is retained outside the final semantic model because its detail-level amounts did not directly reconcile with the bill-level totals. It can potentially be incorporated later if charge-level analysis becomes a requirement and its business rules are clearly established.
- **Do not invent missing dates** — fact_prescription has no prescription date in the source, so prescription analysis cannot legitimately be presented as a prescription-date trend. Similarly, fact_staff_assignment has no assignment date, so historical staff-assignment trends cannot be inferred from this model.
- **Understand the insurance limitation** — fact_patient_insurance can be used for insurance policies, insured patients, insurance providers, coverage percentage and policy dates. However, there is no reliable source relationship linking an individual bill/admission to a specific insurance policy — analysts should not assume a billing transaction belongs to a particular policy.
- **Understand the inventory limitation** — fact_drug_inventory represents current inventory information: current stock, reorder level, inventory status and last restock date. It does not provide historical stock movement.
- **Maintain the flag/junk dimensions** — Some flag/junk dimensions were created using surrogate keys representing combinations of attributes. As new source data enters the warehouse, new combinations may appear; if these are not added to the corresponding flag/junk tables, filters may not work correctly, records may not map correctly, and reports may produce incomplete or incorrect numbers.

This is an important maintenance responsibility when new data is loaded into the warehouse.

Quick Decision Framework and Analyst Checklist

The following decision framework can be used whenever a new analysis is required:

Business Question → Process → Grain → Fact / Dimension → Standard Measure → Date → Filters → Check Relationship Path → DAX if Needed → Validate → Build

Before Publishing a KPI or Dashboard

- ☐ I understand the overall data model.
- ☐ I understand the grain of the relevant table.
- ☐ I identified the correct business process.
- ☐ I selected the appropriate fact/dimension.
- ☐ I used the standard KPI where available.
- ☐ I selected the correct date and date role.
- ☐ I checked cardinality and filter direction where relevant.
- ☐ I confirmed that the required filter path exists.
- ☐ I have not created an unintended fact-to-fact links.
- ☐ I have not assumed every slicer should affect every KPI.
- ☐ I have not invented information that does not exist in the source.
- ☐ I validated the result against the intended business definition.
- ☐ If new source data was loaded, I checked the flag/junk dimensions for new combinations.

FINAL USAGE PRINCIPLE

Read and understand the data model first. Then follow the usage rules in this manual to select the appropriate fact, dimension, measure, date and filter path. Finally, validate the result against the business question before using it in a dashboard or decision.